

On Fault Tolerance Of $D_{3,2,1}$ -Domination*

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Abstract

Distance two domination has practical usage in resource sharing problem. $D_{3,2,1}$ -domination problem is a variation of distance two domination. For a graph $G=(V,E)$, let $D \subseteq V$. Assume that every vertex v in D can get resource completely and share partial resource with its neighbors and offer little resource to the vertices that are distance two from it. Then the weight function of a vertex $v \in V$ may be defined as $w_D(v) = 3|\{v\} \cap D| + 2|N_1(v) \cap D| + |N_2(v) \cap D|$ respect to D where $N_i(v)$ denoted distance i neighborhood of v . $D \subseteq V$ is a $D_{3,2,1}$ -dominating set of G if and only if $w_D(v) \geq 3$ for every vertex $v \in V$. The $D_{3,2,1}$ -domination number $\gamma_{3,2,1}(G)$ of a graph G is the minimum cardinality of a $D_{3,2,1}$ -dominating set of G . The $D_{3,2,1}$ -domination problem is the problem of finding the $D_{3,2,1}$ -domination number of graphs.

This paper corrected the $D_{3,2,1}$ -domination number of double loop graphs in some cases. Due to the possible connection fault, this paper also discussed the edge-fault-tolerance problem respect to $D_{3,2,1}$ -domination on a number of graphs whose $\gamma_{3,2,1}(G)$ are known.

1 Introduction

The domination can be used to model many location problems in operation research. In a graph $G=(V,E)$, a *dominating set* is a subset D of V such that every vertex in $V-D$ is adjacent to at least one vertex in D . The related distance k domination may be used to solve the resource sharing problem. A *distance- k*

dominating set D is a set of vertices such that for every vertex v in $V-D$, there is a vertex at distance at most k from it. $D_{3,2,1}$ -domination is a variation of distance-two domination. For any subset $D \subseteq V$, define the weight-function $w_D : V \rightarrow \mathbb{N}$ as $w_D(v) = 3|\{v\} \cap D| + 2|N_1(v) \cap D| + |N_2(v) \cap D|$ where $N_i(v)$ denoted distance i neighborhood of v , then D is a $D_{3,2,1}$ -dominating set of G if and only if $w_D(v) \geq 3$ for every vertex $v \in V$. The $D_{3,2,1}$ -dominating number of G , denoted as $\gamma_{3,2,1}(G)$, is the minimum cardinality of a $D_{3,2,1}$ -dominating set of G . A $D_{3,2,1}$ -dominating set D of G is *optimal* if and only if $|D| = \gamma_{3,2,1}(G)$. Besides resource sharing, $D_{3,2,1}$ -domination can also be applied to many other problems related to social relationship, including classroom order management, election running and so on. Distance domination has been studied for years and got many results (see [1], [2], [3], [4], [6] [7] [8] [9]). [12] first considered the decreasing effect on distance in 2006. They discussed the $D_{3,2,1}$ -domination problem of paths, cycles and complete full binary trees. In 2008, [11] discussed the $D_{3,2,1}$ -domination problem and its application of double loop networks $DL(n; 1, 2)$ and $DL(n; 1, 3)$. Later on, [5] established the $D_{3,2,1}$ -domination number for the composition graphs of a path with a path and a path with a cycle.

Fault tolerant ability has been discussed respect to many problems including domination [10]. A graph $G=(V,E)$ is said to be k -edge-fault tolerance respect to $D_{3,2,1}$ -domination if for every subset $X \subseteq E$, $|X| \leq k$ such that $\gamma_{3,2,1}(G-X) = \gamma_{3,2,1}(G)$. A graph G is said to be *critical* respect to $D_{3,2,1}$ -domination if and only if for all subset $X \subseteq E$, $X \neq \emptyset$, $\gamma_{3,2,1}(G-X) > \gamma_{3,2,1}(G)$. Motivated by possible

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link errors or communication fault in the applications, this paper discussed the edge fault tolerant ability respect to $D_{3,2,1}$ -domination on a number of graphs including paths, cycles, complete full binary trees, and double loop graphs whose $D_{3,2,1}$ -domination number are known and corrected some errors in previous paper dealt with $D_{3,2,1}$ -domination number of double loop graphs.

2 Paths and Cycles

According to [12], the $D_{3,2,1}$ -domination number of paths and cycles are listed below.

Lemma 1. [12] *The $D_{3,2,1}$ -domination number of path P_n for $n \geq 2$ is $\gamma_{3,2,1}(P_n) = \lfloor (n+4)/3 \rfloor$.*

Lemma 2. [12] *The $D_{3,2,1}$ -domination number of path C_n for $n \geq 3$ can be expressed as*

$$\gamma_{3,2,1}(C_n) = \begin{cases} 2, & \text{if } n = 3; \\ \lceil n/3 \rceil, & \text{if } n > 3. \end{cases}$$

Since removing any edge of a path will result two subpaths. Let P_t and P_k such that $t+k=n$ be two subpaths for some $t, k \geq 1$ obtained by removing an edge from path P_n . Theorem 1 shows that P_n is critical respect to $D_{3,2,1}$ -domination except in the case that both t and k are 1 module 3.

Theorem 1. *Let $P_n = P_t \cup P_k \cup \{e\}$ for $t+k=n$ and $t, k \geq 1$. Then P_n is one-fault tolerance respect to $D_{3,2,1}$ -domination if and only if both t and k are both 1 module 3.*

Proof: Since each faulty-edge will divide the path into subpaths P_t and P_k such that $t+k=n$ for some $t, k \geq 1$, we consider the following cases.

Case 1. One of t and k is 0 module 3, without loss of generality, let $t = 0 \bmod 3$. Then by lemma 1, we have

$$\begin{aligned} \gamma_{3,2,1}(P_t) + \gamma_{3,2,1}(P_k) &= \lfloor (t+4)/3 \rfloor + \lfloor (k+4)/3 \rfloor \\ &= t/3 + 1 + \lfloor (k+4)/3 \rfloor \\ &= 1 + \lfloor (n+4)/3 \rfloor \\ &> \gamma_{3,2,1}(P_n) \end{aligned}$$

Hence in this case P_n is critical.

Case 2. One of t and k is 2 module 3, without loss of generality, let $t = 2 \bmod 3$. Then by lemma 1, we have

$$\begin{aligned} \gamma_{3,2,1}(P_t) + \gamma_{3,2,1}(P_k) &= \lfloor (t+4)/3 \rfloor + \lfloor (k+4)/3 \rfloor \\ &= (t-2)/3 + 2 + \lfloor (k+4)/3 \rfloor \end{aligned}$$

Since $(t-2)/3$ is an integer, we have

$$\begin{aligned} \gamma_{3,2,1}(P_t) + \gamma_{3,2,1}(P_k) &= 2 + \lfloor (n+2)/3 \rfloor \\ &= 1 + \lfloor (n+5)/3 \rfloor \\ &> \lfloor (n+4)/3 \rfloor = \gamma_{3,2,1}(P_n) \end{aligned}$$

Hence in this case P_n is critical.

Case 3. Both t and k are 1 module 3. In this case, $n = 2 \bmod 3$. Then by lemma 1, we have

$$\begin{aligned} \gamma_{3,2,1}(P_n) &= \lfloor (n+4)/3 \rfloor = \lfloor (t+k+4)/3 \rfloor \\ &= (t-1)/3 + (k-1)/3 + 2 \\ &= \lfloor t/3 \rfloor + \lfloor k/3 \rfloor + 2 \\ &= \lfloor t/3 + 1 \rfloor + \lfloor k/3 + 1 \rfloor \\ &= \lfloor (t+4)/3 \rfloor + \lfloor (k+4)/3 \rfloor \\ &= \gamma_{3,2,1}(P_t) + \gamma_{3,2,1}(P_k) \end{aligned}$$

Hence P_n is one-fault tolerance in this case. $\square$

Since removing any edge of a cycle C_n will result a path P_n , Theorem 2 established that C_n is critical except when $n = 3k+1$ for some integer k in which case C_n is one-fault tolerance.

Theorem 2. *C_n is one-fault tolerance if and only if $n = 3k+1$ for some integer k .*

Proof: Consider the following cases.

Case 1. $n = 3k$ for some integer k . Then by lemma 1 and lemma 2, we have

$$\begin{aligned} \gamma_{3,2,1}(P_n) &= \lfloor (n+4)/3 \rfloor = k+1 \\ &> k = \lceil n/3 \rceil = \gamma_{3,2,1}(C_n) \end{aligned}$$

Hence C_n is critical in this case.

Case 2. $n = 3k+2$ for some integer k . Then by lemma 1 and lemma 2, we have

$$\begin{aligned} \gamma_{3,2,1}(P_n) &= \lfloor (n+4)/3 \rfloor = k+2 \\ &> k+1 = \lceil n/3 \rceil = \gamma_{3,2,1}(C_n) \end{aligned}$$

Hence C_n is critical in this case .

Case 3. $n = 3k + 1$ for some integer k . Then by lemma 1 and lemma 2, we have

$$\begin{aligned}\gamma_{3,2,1}(P_n) &= \lfloor (n+4)/3 \rfloor = k+1 \\ &= \lceil n/3 \rceil = \gamma_{3,2,1}(C_n)\end{aligned}$$

Hence C_n is one-fault tolerance in this case. $\square$

3 Complete full binary trees

A complete full binary tree B_n is a rooted tree with height n and each internal vertex has exactly two children. The $D_{3,2,1}$ -domination number of B_n given by [12] is listed as lemma 3.

Lemma 3. [12] The $D_{3,2,1}$ -domination number of a complete full binary tree B_n with height n is

$$\gamma_{3,2,1}(B_n) = \begin{cases} 1 + \sum_{j \in S} 2^j, & n = 5i, i \geq 0; \\ 2 + \sum_{j \in S'} 2^j, & n = 1 + 5i, i \geq 0; \\ \sum_{j \in S} 2^j, & \text{otherwise.} \end{cases}$$

where $S = \{n-2-5t, n-1-5t\}$ for $0 \leq t \leq \lfloor (n+4)/5 \rfloor - 1$, and $S' = \{n-2-5t, n-1-5t\}$ for $0 \leq t \leq \lfloor (n+4)/5 \rfloor - 2$.

According to lemma 3, the $D_{3,2,1}$ -dominating set of B_n must contain all of the vertices in level j where $j \in S'$ for $n = 5i + 1$ and $j \in S$ otherwise. Next theorem shows that although B_n is zero-fault tolerance in general but B_n is not critical respect to $D_{3,2,1}$ -domination. For the convenience, let $l(v)$, $r(v)$ and $p(v)$ denote the left-child, right-child and the parent of vertex v respectively. Let $T(v)$ denote the tree rooted by v ; $T_l(v)$ and $T_r(v)$ denote the left subtree and right subtree rooted by $l(v)$ and $r(v)$ respectively. Figure 1 shows the relation of a B_n with these notations.

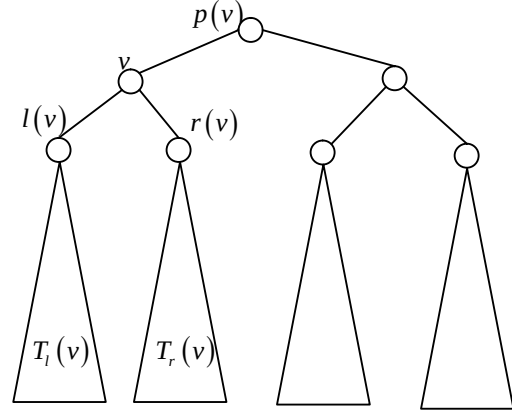


Figure 1: Relation of notation in B_n

Lemma 4. Let D be an optimal $D_{3,2,1}$ -dominating set of a complete full binary tree B_n for $n \geq 2$. Then D does not contain any leaf vertex of B_n .

Proof: Without lose of generality, assume to the contrary that there is a leaf vertex $l(v) \in D$ and v is a left-child of its parent. Since another leaf vertex $r(v)$ has only two distance-two neighbors, either v or $r(v)$ has to be in D . Hence $|T_l(p(v)) \cap D| \geq 2$. Since the leaves in $T_r(p(v))$ does not get any weight from $T_l(p(v))$, they request at least two vertices of $T_r(p(v)) \cup p(v)$ in D . Hence $|T(p(v)) \cap D| \geq 4$. Now consider $D' = (D - T(p(v))) \cup \{p(v), v, r(p(v))\}$, notice that $|D'| < |D| = \gamma_{3,2,1}(B_n)$. Since every leaves of $T(p(v))$ are distance two from $p(v)$ and distance one from either v or $r(p(v))$, D' is a $D_{3,2,1}$ -dominating set of B_n which contradict to D is optimal. $\square$

Theorem 3. The complete full binary tree B_n with height $n \geq 2$ has zero-fault tolerance and B_n is critical respect to $D_{3,2,1}$ -domination if and only if $n = 2$.

Proof. We first show that B_n has zero-fault tolerance. Let D be an optimal $D_{3,2,1}$ -dominating set of B_n . By lemma 4, D does not contain any leaf. Let $l(v)$ be a leaf of B_n and $e_1 = (v, l(v))$. Then $\gamma_{3,2,1}(B_n - e_1) =$

$\gamma_{3,2,1}(B_n)+1$. Hence B_n has zero-fault tolerance. Next we show that B_n is not critical for $n \geq 3$. Let $e_2 = (p(v), p(p(v)))$ for some leaf $l(v)$. Then $\gamma_{3,2,1}(B_n - e_2) = \gamma_{3,2,1}(B_n)$. Hence B_n is not critical respect to $D_{3,2,1}$ -domination for $n \geq 3$. For $n = 2$, since every edge is a bridge in a tree, for any edge $e \in E(B_2)$, $B_2 - e$ has two components H_1 and H_2 . If both H_i are not trivial, then $\gamma_{3,2,1}(H_i) \geq 2$ for $i = 1, 2$. Hence $\gamma_{3,2,1}(B_2 - e) \geq 2 + 2 = 4 > 3 = \gamma_{3,2,1}(B_2)$. If one of H_i is trivial, (say H_1), then $\gamma_{3,2,1}(H_1) = 1$ and $\gamma_{3,2,1}(H_2) = \gamma_{3,2,1}(B_2) = 3$. Hence $\gamma_{3,2,1}(B_2 - e) = 1 + 3 = 4 > 3 = \gamma_{3,2,1}(B_2)$. In both cases we have B_2 is critical respect to $D_{3,2,1}$ -domination. $\square$

In fact, even the faulty-edge is not a pending edge in a complete full binary tree, B_n still has no fault tolerant ability. By lemma 3, the optimal $D_{3,2,1}$ -dominating set of B_n must contain all vertices of some pairs of consecutive levels, any faulty-edge e between those two consecutive levels in the optimal $D_{3,2,1}$ -dominating set will have to increase the $D_{3,2,1}$ -domination number in the resulting graph.

4 Double Loop Graphs

A double loop graph denoted as $DL(n; a, b)$ in this paper is an underlining graph of a directed graph with n vertices $0, 1, 2, \dots, n-1$ and $2n$ directed edges of the form $i \rightarrow i+a \pmod{n}$ and $i \rightarrow i+b \pmod{n}$, referred to as a -links and b -links. In 2008, [6] established the $D_{3,2,1}$ -domination number of $DL(n; 1, 2)$ which we state as lemma 5. They also provided the $D_{3,2,1}$ -domination number of $DL(n; 1, 3)$, however, there are some small errors which we corrected it as lemma 6.

Lemma 5. [11] *Let $G = DL(n; 1, 2)$ be a double loop graph of order $n \geq 4$, then the $D_{3,2,1}$ -domination number of G is*

$$\gamma_{3,2,1}(G) = \begin{cases} 2, & \text{if } n = 4 \text{ or } 5; \\ \lceil n/5 \rceil, & \text{if } n > 5. \end{cases}$$

The original theorem for the $D_{3,2,1}$ -domination

number of $DL(n; 1, 3)$ in [6] is for any n , $\gamma_{3,2,1}(DL(n; 1, 3)) = \lceil n/5 \rceil$, which we corrected as lemma 6.

Lemma 6. *Let $G = DL(n; 1, 3)$ be a double loop graph of order $n \geq 6$, then the $D_{3,2,1}$ -domination number of G is*

$$\gamma_{3,2,1}(G) = \begin{cases} \lceil n/5 \rceil + 1, & \text{if } n = 9 \text{ or } 14; \\ \lceil n/5 \rceil, & \text{otherwise.} \end{cases}$$

Proof: Let $G_1 = DL(9; 1, 3)$ and $G_2 = DL(14; 1, 3)$. Then we will show that $\gamma_{3,2,1}(G_1) = 3$ and $\gamma_{3,2,1}(G_2) = 4$. Since it is easy to see that $\{0, 5, 7\}$ and $\{0, 5, 7, 9\}$ are $D_{3,2,1}$ -dominating set of G_1 and G_2 respectively, which implies $\gamma_{3,2,1}(G_1) \leq 3$ and $\gamma_{3,2,1}(G_2) \leq 4$. To see the lower bound, we consider following two cases:

Case 1: $n = 9$. Since G_1 is a 4-regular graph, every vertex has four distance-two neighbors which form a path P_4 . Since every vertex in G_1 is adjacent to at most three of these four vertices, hence $\gamma_{3,2,1}(G_1) > 2$.

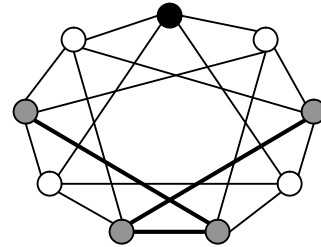


Figure 2: The four vertices get weight 1 from v

Case 2: $n = 14$. Let D be a $D_{3,2,1}$ -dominating set of G_2 . Without loss of generality, let vertex $0 \in D$. There are six possible subcases which we state as follows.

Case 2.1: either vertex $1 \in D$ or vertex $3 \in D$. In this case there are six vertices have only get weight one from $\{0, 1\}$ or $\{0, 3\}$. Since G_2 is a 4-regular graph, there is no vertex is within distance one of these six vertices. Hence we have $|D| > 3$.

Case 2.2: vertex $2 \in D$. In this case, there are two vertices do not get any weight from $\{0, 2\}$ which requires at least two more vertices in D to

$D_{3,2,1}$ -dominate those two vertices. Hence we have $|D| > 3$.

Case 2.3: vertex $4 \in D$. Since vertex 9 is distance three from both vertices 0 and 4, that implies vertex $9 \in D$. Consider $D' = \{0, 4, 9\} \subseteq D$. Since vertex 2 is distance two from both vertices 0 and 4, and distance three from vertex 9, we have $w_{D'}(2) = 2$. Since D is a $D_{3,2,1}$ -dominating set of G_2 and $D' \subseteq D$ is not a $D_{3,2,1}$ -dominating set of G_2 , we have $|D| > 3$.

Case 2.4: vertex $5 \in D$. In this case, the vertices 7, 9, 10, and 12 only get weight 1 from $\{0, 5\}$, and there is no vertex in G_2 are within distance one from all four of them. Hence we have $|D| > 3$.

Case 2.5: vertex $6 \in D$. In this case, every vertex gets weight 2 from $\{0, 6\}$. Since each vertex in G_2 has only four distance-one neighbors and six distance-two neighbors, a vertex can offer some weight to at most 11 vertices (containing itself). Since there are 12 vertices still have one short for weight, we have $|D| > 3$.

Case 2.6: vertex $7 \in D$. In this case, the four vertices getting only one weight from $\{0, 7\}$ are vertices 2, 5, 9, and 12. Since there is no vertex in G_2 are within distance one from all four of them, we have $|D| > 3$. $\square$

Next two theorems show the fault-tolerant ability of the graph $DL(n; 1, 2)$ and $DL(n; 1, 3)$.

Theorem 4 A double loop graph $DL(n; 1, 2)$ is one-fault tolerance.

Proof. To show that $DL(n; 1, 2)$ is one-fault tolerance, we consider following two cases according to the type faulty edge.

Case 1: The faulty edge is a 1-link. Without loss of generality, let $(1, 2)$ be the faulty edge of $DL(n; 1, 2)$. Then $D = \{5i \mid 0 \leq i \leq (n-1)/5\}$ is a $D_{3,2,1}$ -dominating set of $DL(n; 1, 2) - (1, 2)$ and $|D| = \gamma_{3,2,1}(DL(n; 1, 2))$ which implies that $DL(n; 1, 2)$ is one-fault tolerance in this case.

Case 2: The faulty edge is 2-link. Without loss of generality, let $(1, 3)$ be the faulty edge of

$DL(n; 1, 2)$. Then $D = \{5i \mid 0 \leq i \leq (n-1)/5\}$ is a $D_{3,2,1}$ -dominating set of $DL(n; 1, 2) - (1, 3)$ and $|D| = \gamma_{3,2,1}(DL(n; 1, 2))$ which implies that $DL(n; 1, 2)$ is one-fault tolerance in this case. $\square$

In fact, $DL(n; 1, 2)$ is not only one-fault tolerance, but has conditional two-fault tolerance. As long as the faulty edges are not adjacent to the same vertex in the same direction, that is, every vertex has at least one good edge from each side of the cycle, we then are able to use the same number of vertices as $\gamma_{3,2,1}(DL(n; 1, 2))$ to $D_{3,2,1}$ -dominate the resulting graph.

Corollary 1. A double loop graph $DL(n; 1, 2)$ has two-fault tolerance with condition that the faulty-edges are not joined to the same vertex in the same direction.

Theorem 5 A double loop graph $DL(n; 1, 3)$ is one-fault tolerance.

Proof: To show that $DL(n; 1, 3)$ is one-fault tolerance, we consider following cases according to the order of graph.

Case 1: $n \neq 5k + 4$ for some integer $k \geq 3$. Since the faulty edge may be either a 1-link or a 3-link, without loss of generality, let $(1, 2)$ or $(1, 4)$ be the faulty edge of $DL(n; 1, 3)$. Consider the optimal $D_{3,2,1}$ -dominating set $D = \{5i \mid 0 \leq i \leq \lceil n/5 \rceil - 1\}$ of $DL(n; 1, 3)$ for $n \neq 5k + 4$. We show that D is also a $D_{3,2,1}$ -dominating set of $DL(n; 1, 3) - (1, 2)$ and $DL(n; 1, 3) - (1, 4)$. Notice that the vertices affected by the faulty edges are vertices $\{1, 2, 4\}$. Since without one of two edges $(1, 2)$ and $(1, 4)$, vertex 1 is distance one from vertex 0 and distance two from vertex 5. The distance between vertices 2 and 4 with vertices 0 and 5 is not changed. Hence D is also a $D_{3,2,1}$ -dominating set of $DL(n; 1, 3) - (1, 2)$ and $DL(n; 1, 3) - (1, 4)$, which implies that $DL(n; 1, 3)$ is one-fault tolerance in this case.

Case 2. $n = 5k + 4$, and $k \geq 3$. Without loss of generality, let either $(9, 10)$ or $(9, 12)$ be the faulty edge of $DL(n; 1, 3)$. Consider the optimal $D_{3,2,1}$ -dominating set $D = \{0, 2\} \cup \{5i + 3 \mid 1 \leq i \leq \lfloor n/5 \rfloor - 1\}$ of $DL(n; 1, 3)$ for $n = 5k + 4$,

$k \geq 3$. We want to show that D is also a $D_{3,2,1}$ -dominating set of $DL(n; 1, 3) - (9, 10)$ and $DL(n; 1, 3) - (9, 12)$. Notice that the vertices affected by the faulty edges are vertices $\{9, 10, 12\}$. Since without one of two edges $(9, 10)$ and $(9, 12)$, vertex 9 is distance one from vertex 8 and distance two from vertex 13. The distance between vertices 10 and 12 with vertices 8 and 13 is not changed. Hence D is also a $D_{3,2,1}$ -dominating set of $DL(n; 1, 3) - (9, 10)$ and $DL(n; 1, 3) - (9, 12)$, which implies that $DL(n; 1, 3)$ is one-fault tolerance in this case.

Case 3. $n = 5k + 4$ and $1 \leq k \leq 2$. Without loss of generality, let either $(1, 2)$ or $(1, 4)$ be the faulty edge of $DL(n; 1, 3)$. Then consider the optimal $D_{3,2,1}$ -dominating set $D = \{n - 2\} \cup \{5i \mid 0 \leq i \leq \lceil n/5 \rceil - 1\}$ of $DL(n; 1, 3)$, similar to case 1, D is also a $D_{3,2,1}$ -dominating set of $DL(n; 1, 3) - (1, 2)$ and $DL(n; 1, 3) - (1, 4)$. Hence $DL(n; 1, 3)$ is one-fault tolerance. $\square$

5 Conclusion

By giving different weight to each vertex according to the distance in the domination set, $D_{3,2,1}$ -domination can not only represent the effect decreasing of the shared resource caused by distance but can also represent the decreasing of the influence on people by their close relationship. Since the edges represent the direct link or close relation between people, it is reasonable to consider the edge fault tolerance ability respect to $D_{3,2,1}$ -domination. This paper shows that paths, and cycles have one-fault tolerance under certain condition and are critical for other cases. For complete full binary trees B_n with height n , they all have zero-fault tolerance but only B_2 is critical. For double loop graphs, we corrected the $D_{3,2,1}$ -dominating number given by [11] for $DL(9; 1, 3)$ and $DL(14; 1, 3)$ and proved that both $DL(n; 1, 2)$ and $DL(n; 1, 3)$ have one-fault tolerance respect to $D_{3,2,1}$ -domination.

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